



# Cambridge IGCSE™

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**MATHEMATICS**

**0580/23**

Paper 2 (Extended)

**October/November 2021**

**MARK SCHEME**

Maximum Mark: 70

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**Published**

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2021 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

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This document consists of **6** printed pages.

**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

**GENERIC MARKING PRINCIPLE 1:**

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

**GENERIC MARKING PRINCIPLE 2:**

Marks awarded are always **whole marks** (not half marks, or other fractions).

**GENERIC MARKING PRINCIPLE 3:**

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

**GENERIC MARKING PRINCIPLE 4:**

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

**GENERIC MARKING PRINCIPLE 5:**

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

**GENERIC MARKING PRINCIPLE 6:**

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

| Maths-Specific Marking Principles |                                                                                                                                                                                                                                               |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                                 | Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.                                     |
| 2                                 | Unless specified in the question, answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.                                                            |
| 3                                 | Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.                                                                                                               |
| 4                                 | Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).                                                                                            |
| 5                                 | Where a candidate has misread a number in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 mark for the misread. |
| 6                                 | Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.                                                                                              |

**Abbreviations**

|     |                            |
|-----|----------------------------|
| cao | correct answer only        |
| dep | dependent                  |
| FT  | follow through after error |
| isw | ignore subsequent working  |
| oe  | or equivalent              |
| SC  | Special Case               |
| nfw | not from wrong working     |
| soi | seen or implied            |

| Question | Answer                                                                                | Marks     | Partial Marks                                                                                                                                                                                                   |
|----------|---------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | 12.5                                                                                  | 1         |                                                                                                                                                                                                                 |
| 2        | 48                                                                                    | 2         | <b>B1</b> for 132 or 48 in the correct position on the diagram<br>or<br><b>M1</b> for $180 - 132$                                                                                                               |
| 3        | 13                                                                                    | 1         |                                                                                                                                                                                                                 |
| 4(a)     | $\neq$ and $>$ indicated                                                              | 1         |                                                                                                                                                                                                                 |
| 4(b)     | $7 - (3 - 1) + 2 = 7$ cao                                                             | 1         |                                                                                                                                                                                                                 |
| 5        | 170                                                                                   | 1         |                                                                                                                                                                                                                 |
| 6        | 357                                                                                   | 2         | <b>M1</b> for $\left(1 - \frac{15}{100}\right) \times 420$ oe<br>or <b>B1</b> for 63                                                                                                                            |
| 7        | $8g^{28}$ final answer                                                                | 2         | <b>B1</b> for $kg^{28}$ or $8g^k$ as final answer or correct answer seen and spoilt                                                                                                                             |
| 8        | 4.32                                                                                  | 3         | <b>B1</b> for $\frac{1}{4}$ oe or $\frac{2}{4.5}$ oe seen<br><b>M1 dep</b> on <b>B1</b> for $\frac{1+2}{their \frac{1}{4} + their \frac{2}{4.5}}$ oe                                                            |
| 9        | $\frac{3}{11}$ oe fraction                                                            | 1         |                                                                                                                                                                                                                 |
| 10(a)    | -13                                                                                   | 1         |                                                                                                                                                                                                                 |
| 10(b)    | $-4n + 7$ oe final answer                                                             | 2         | <b>B1</b> for $-4n + k$ or $jn + 7$ ( $j \neq 0$ ) or for a correct answer spoilt                                                                                                                               |
| 11(a)    | 2925                                                                                  | 2         | <b>M1</b> for $100(3^2 + 4.5^2)$<br>or <b>B1</b> for 29.25 seen                                                                                                                                                 |
| 11(b)    | $[\pm] \sqrt{\frac{P}{M} - h^2}$<br>or $[\pm] \sqrt{\frac{P - Mh^2}{M}}$ final answer | 3         | <b>M1</b> for correct division by $M$<br><b>M1</b> for correct re-arrangement to isolate $g$ or $g^2$<br><b>M1</b> for correct square root of two term expression<br>Max <b>2</b> marks for an incorrect answer |
| 12       | $\frac{11}{12} + \frac{9}{12}$ oe                                                     | <b>M1</b> | Allow any correct common denominator $12k$                                                                                                                                                                      |
|          | $1\frac{2}{3}$ cao                                                                    | <b>A2</b> | <b>A1</b> for $\frac{20}{12}$ or equivalent improper fraction or mixed number                                                                                                                                   |

| Question | Answer                        | Marks | Partial Marks                                                                                                                                                                                                                                                                                                                                                                              |
|----------|-------------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13       | $1[.0] \times 10^{-2}$ cao    | 2     | <b>B1</b> for 0.01 oe                                                                                                                                                                                                                                                                                                                                                                      |
| 14(a)    | $b, c, d, e, f, g$            | 1     |                                                                                                                                                                                                                                                                                                                                                                                            |
| 14(b)    | 4                             | 1     |                                                                                                                                                                                                                                                                                                                                                                                            |
| 14(c)    | 3                             | 1     |                                                                                                                                                                                                                                                                                                                                                                                            |
| 15       | 145                           | 2     | <b>M1</b> for $x\left(1 + \frac{6}{100}\right) = 153.7$ oe or better                                                                                                                                                                                                                                                                                                                       |
| 16       | 31 : 21                       | 3     | <b>B2</b> for equivalents e.g. 15.5 oe and 10.5 oe or for an equivalent ratio e.g. 3.1 : 2.1<br><br>or <b>M1</b> for e.g. $x + 5 + x = 26$ oe or $x - 5 + x = 26$ oe                                                                                                                                                                                                                       |
| 17       | 240                           | 2     | <b>M1</b> for $360 \div (180 - 178.5)$ oe<br>or for $\frac{180(n-2)}{n} = 178.5$ oe                                                                                                                                                                                                                                                                                                        |
| 18       | $[y =] 12x - 26$ final answer | 3     | <b>M1</b> for $\frac{10 - -2}{3 - 2}$ oe<br><br><b>M1</b> for correct substitution of (2, -2) or (3, 10)<br>into $y = (their\ m)x + c$ oe                                                                                                                                                                                                                                                  |
| 19       | 33.8 or 33.78 to 33.80        | 4     | <b>M2</b> for $2 \times 12.6 \times \sin 40$ oe<br>or <b>M1</b> for $\sin 40 = \frac{(\dots)}{12.6}$ oe<br><br><b>M1</b> for $\frac{80}{360} \times 2 \times \pi \times 12.6$ oe                                                                                                                                                                                                           |
| 20       | 40 000                        | 3     | <b>B2</b> for 1 cm to 0.4 km or 2.5 cm to 1 km or 1 600 000 000<br><br>or <b>M2</b> for $\sqrt{\frac{3 \times 10^k}{18.75}}$ oe where $k > 5$<br><br>or <b>M1</b> for 1 cm <sup>2</sup> to 0.16 km <sup>2</sup> or 6.25 cm <sup>2</sup> to 1 km <sup>2</sup><br>or for $3 \times 10^{10}$ oe<br>or $1.875 \times 10^{-9}$ oe<br>or $3 \times 10^6$ oe <b>and</b> $1.875 \times 10^{-3}$ oe |
| 21       | $27y^6$ final answer          | 2     | <b>B1</b> for $ky^6$ or $27y^k$ as final answer or correct answer seen and spoilt                                                                                                                                                                                                                                                                                                          |

| Question | Answer                                                                                      | Marks     | Partial Marks                                                                                                                                                                                                                                                                                                                                                                                             |
|----------|---------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 22       | $x^2 - 4x - 12 [= 0]$<br>or<br>$y^2 - 2y - 15 [= 0]$                                        | <b>M2</b> | <b>M1</b> for $x^2 - 3x - 13 = x - 1$<br>or for $y = (y + 1)^2 - 3(y + 1) - 13$                                                                                                                                                                                                                                                                                                                           |
|          | $(x - 6)(x + 2) [= 0]$<br>or<br>$(y - 5)(y + 3) [= 0]$                                      | <b>M1</b> | or for correct factors for <i>their</i> quadratic equation<br>or for correct use of quadratic formula or completing the square for <i>their</i> equation                                                                                                                                                                                                                                                  |
|          | $[x =] 6, [y =] 5$<br>$[x =] -2, [y =] -3$                                                  | <b>B2</b> | <b>B1</b> for one correct pair or two correct $x$ values or two correct $y$ values<br><br>If <b>B0</b> scored <b>and</b> at least 2 method marks scored <b>SC1</b> for correct substitution of both of <i>their</i> $x$ values or <i>their</i> $y$ values into $y = x^2 - 3x - 13$ or $y = x - 1$                                                                                                         |
| 23(a)    | 13.6 or 13.60...                                                                            | <b>3</b>  | <b>M2</b> for $12^2 + 5^2 + 4^2$<br>or <b>M1</b> for $5^2 + 4^2$ or $12^2 + 4^2$ or $12^2 + 5^2$                                                                                                                                                                                                                                                                                                          |
| 23(b)    | 17.1 or 17.08 to 17.10...                                                                   | <b>3</b>  | <b>M2</b> for $\sin = \frac{4}{\text{their (a)}}$ oe or<br>$\tan = \frac{4}{\text{their AP}}$ or $\cos = \frac{\text{their AP}}{\text{their (a)}}$<br>or <b>M1</b> for recognising angle <i>CAP</i> .                                                                                                                                                                                                     |
| 24       | 60<br>and<br>240                                                                            | <b>2</b>  | <b>B1</b> for 60 or 240<br><br>If 0 scored <b>SC1</b> for two answers with a difference of $180^\circ$                                                                                                                                                                                                                                                                                                    |
| 25       | $\frac{3x}{a + 2c}$ final answer                                                            | <b>4</b>  | <b>B1</b> for $3x(x - 6)$<br><b>B2</b> for $(x - 6)(a + 2c)$<br>or <b>B1</b> for $a(x - 6) + 2c(x - 6)$ or $x(a + 2c) - 6(a + 2c)$                                                                                                                                                                                                                                                                        |
| 26       | $\frac{3}{5}\mathbf{r} + \frac{2}{5}\mathbf{t}$ or $\frac{1}{5}(3\mathbf{r} + 2\mathbf{t})$ | <b>3</b>  | <b>M2</b> for $\mathbf{r} + \frac{2}{5}(-\mathbf{r} + \mathbf{t})$ oe or $\mathbf{t} + \frac{3}{5}(\mathbf{r} - \mathbf{t})$<br>oe<br>or<br><b>M1</b> for $\overrightarrow{RT} = -\mathbf{r} + \mathbf{t}$ oe or $\overrightarrow{TR} = \mathbf{r} - \mathbf{t}$<br><br><b>M1</b> for $\overrightarrow{OR} + \overrightarrow{RX}$ or $\overrightarrow{OT} + \overrightarrow{TX}$ any other correct route. |